Software Requirements Specification

Jobinator

Version 1.0

Prepared by: {Ethan, Brian, Mohammad, Kevin}

4/8/26

Reserved

Table Of Contents

1. Introduction 5
   1.1 Software Purpose 5
   1.2 System Overview 5
   1.3 Project Intent 6
   1.4 Current Release 6
   1.5 Future Release 6
2. Overall Description 7
   2.1 Product Overview 7
   2.2 Operating Environment 8
      1. Hardware Platform 8
      2. Operating Systems and Device Support 8
      3. Network Environment 9
      4. Data Storage Environment 9
      5. Other Software Attributes 9
   2.3 Design and Implementation Constraints 10
      1. Platform and Tech Constraints 10
      2. Performance Constraints 10
      3. AI Output Limitations 11
      4. Security and Privacy Constraints 11
      5. Legal and Ethical Constraints 11
      6. UI/UX Constraints 12
      7. Development Process Constraints 12
   2.4 Assumptions and Dependencies 12
      1. AI Output Accuracy and Relevance 12
      2. External AI Service Availability 13
      3. User Input Completeness and Accuracy 13
      4. File Upload and Reference Processing 14
      5. Privacy, Security, and Permission Handling 14
      6. System Performance Under Load 14
      7. Browser, Device, and Interface Compatibility 15
      8. Data Persistence and Export Reliability 15

Reserved

3. System Features 16
3.1 Functional Requirements 16
1. User Account Management 16
2. Document Creation and Input Management 16
3. AI Content Generation and Improvements 17
4. Job-Specific Tailoring 17
5. Document Saving, Editing, and Exporting 17
6. Feedback and Suggestions 18
7. Admin Monitoring and Management 18
8. Privacy and Permissions 18
3.2 Use Case Diagram 19
3.3 Use Case Description 20
4. Data Requirements 23
4.1 Entity Relationship Diagram 23
4.2 Data Flow Diagram 24
4.3 Mockup Design 24
1. Login View 25
2. Dashboard View 26
3. Create View 28
4. Input View 29
5. Editor with AI suggestions 31
6. Tailor Document View 33
7. Export View 34
5. External Interface Requirements 36
5.1 User Interfaces 36
1. Web-Based User Interface 36
2. Responsive Design 36
3. Document Input Interface 36
4. Guided Question Interface 36
5. Document Editor Interface 36
6. Tailoring Interface 37
7. Export Interface 37
8. Administrative Interface 37

Reserved

5.2 Software Interfaces 37
1. AI Service Interface 37
2. Backend Application Server 37
3. Database and Storage Interface 37
4. Document Processing Interface 38
5. Authentication Interface 38
6. Reference and URL Handling Interface 38
7. Administrative AI Configuration Interface 38
5.3 Communication Interfaces 38
1. HTTPS Communication 38
2. RESTful API Communication 38
3. JSON Data Exchange 39
4. File Upload Communication 39
5. AI Service Communication 39
6. Authentication and Session Communication 39
7. Admin Communication Pathways 39
6. Quality Attributes 40
6.1 Performance 40
6.2 Usability 40
6.3 Reliability 40
6.4 Security 41
6.5 Scalability 41
6.6 Maintainability 42
6.7 Accessibility 42
6.8 Compatibility 42

Reserved

1. Introduction

Jobinator is a software project focused on assisting users with the creation and improvement of professional career documents. Rather than functioning as a simple text editor, the system itself is expected to act as an AI-assisted platform that helps the users organize their qualifications, strengthen written content, and adapt their materials to professional standards. The project is centered on three career tools: resumes, curriculum vitae (CVs), and LinkedIn profiles.

1.1 Software Purpose

The purpose/goal of this software requirements specification is to define the expected behaviors, scope, and technical direction of Jobinator. This document serves as the reference point for how the system should operate, what services it should provide, and what limitations or constraints must be considered during development.

It is intended to support communication among the development team, designers, instructors, and any stakeholders/investors/investing parties reviewing the project. By documenting the requirements clearly, the team can maintain a shared understanding of the system before moving into implementation, testing, and design refinement.

1.2 System Overview

Jobinator is designed for users who need help building professional self-presentation materials for academic, internship, or job-seeking purposes. The system supports two general situations:

- A user may begin with little or no existing material and use the platform to generate a resume, CV, or LinkedIn profile from scratch.
- A user may already have an existing resume, CV, or LinkedIn profile and want the system to improve its structure, wording, clarity, or relevance to a specific role. For LinkedIn specifically, the system may allow the user to provide profile content manually with pictures or submit a LinkedIn profile URL as a reference source.

Reserved

1.3 Project Intent

The main intent of Jobinator is to make career document development more accessible, efficient, and personalized. Many users struggle not because they lack experience, but because they do not know how to present that experience professionally. Jobinator addresses that problem by helping users convert rough drafts/raw background information into structured, readable, and role-relevant material.

1.4 Current Release

The current release of Jobinator focuses on delivering the core features needed to help users create, improve, and manage professional career documents. The first version of the platform includes:

- User account creation, secure login, logout, profile updates, and account recovery
- Support for creating or improving a **resume, CV, or LinkedIn profile**
- Manual input of personal, academic, and professional information
- Upload or pasted input for existing resume and CVs
- LinkedIn profile support through pasted content or a LinkedIn profile URL
- AI assisted content generation, rewriting, and wording improvement
- Job specific tailoring based on a target role, industry, or job description.
- Saving, reopening, editing, and exporting documents in formats such as **PDF or DOCX**
- AI generated suggestions for weak sections, missing content, and document refinement
- User controlled privacy settings and permission handling
- Administrative oversight for account management, issue review, and AI service configuration

1.5 Future Release

Future releases of Jobinator may expand the platform beyond the core document building workflow and introduce stronger security, usability, and system refinement features. Planned or possible future improvements include:

- Enhanced document security features for saved work and returning sessions
- A secure re-entry system using account linked verification or recovery keys for access to previously saved documents

Reserved

- Stronger session protection controls to reduce accidental data exposure or unauthorized reuse
- Additional export and formatting options for different professional document styles
- Expanded integrations for external profile or job reference sources
- More advanced AI refinement features for deeper personalization and role targeting
- Improved long term document management, including stronger retention, deletion, and recovery controls.

2. Overall Description

2.1 Product Overview

Jobinator is an AI-assisted career document platform/software created to support users in developing stronger professional materials for academic and mainly employment purposes. The system is centered around three major outputs: Resumes, Curriculum Vitae (CVs), and LinkedIn profile content. Its primary role is to help users transform unformatted/rough-sketch/raw background information into more organized/polished documents.

Rather than having a regular word processor that analyzes and spits out randomized structures, Jobinator is intended to function as an intelligent support tool layered on top of the document-building process. The system assists with structure, wording, and refinement, while still relying on the user to provide their personal, academic, and professional information.

Jobinator will operate as part of a bigger software environment that will include several connected parts/components, such as, for example:

- An AI-based text generation or recommendation engine for drafting and reviewing content
- A user web interface for entering, reviewing, and editing information
- A backend application server for processing requests, storing user-data temporarily and securely, and managing sessions
- Optional document import or export services for handling files such as a PDF or other popular file types
- Optional support for user-provided LinkedIn profile references, such as profile text or profile URLs, along with job description input sources for document refinement/tailoring.

This system is not intended to replace the user's judgement/ideas or automatically guarantee hiring success. Its role is to improve the efficiency, clarity, and professionalism of career documents by offering guided support with an AI during creating or revising the users' documents.

Reserved

2.2 Operating Environment

Jobinator is designed as a web-based software system accessible through standard desktop and mobile browsing environments. The platform is intended to be available through modern browsers and operate across common operating systems without requiring special hardware or other installed applications.

1. Hardware Platform

The system is expected to run on cloud-hosted or server-based infrastructure that supports:

- User authentication and session handling
- Document processing and storage
- AI-driven text generation or rewriting
- Request handling between the client interface and backend services

Stable server resources are still necessary to support multiple users, document uploads, and AI response generation without severe server delay. Because the project focuses on career documents rather than high-intensity media processing, the hardware demands are moderate/medium compared to media-heavy systems like Instagram or TikTok.

2. Operating Systems and Device Support

Jobinator should support access from the major consumer OSs and browsers, including:

- Windows
- MacOS
- Linux
- Android
- iOS

Supported browsers should include:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari
- And other minimal browsers weren't mentioned

The interface should remain usable across desktop, laptop, tablet, and mobile. Needs responsive layout behavior adapted to screen size so it doesn't cause issues/crashes.

Reserved

3. Network Environment

If and when Jobinator relies on server-side processing and AI-assisted features, an internet connection is required for normal use. Network access is especially important when users are:

- Logging into the platform
- Generating or revising document content
- Uploading existing files
- Saving or retrieving profile information
- Interacting with optional external services

Slower connections may increase response time when AI generation or file handling is involved, while low bandwidth shouldn't prevent basic use of it all.

4. Data Storage Environment

The system is expected to store information that the user has entered in a secure backend database or managed cloud storage system. The stored data may have/include:

- Account information
- Education history
- Work experience
- Skills and certifications
- AI-generated revisions
- Saved complete or uncomplete rough drafts
- The most important: user preferences, choices, or customization settings

An organized/structured database system would likely be used for account and document data, while file storage structures may be used for uploaded resumes or exported versions.

5. Other Software Attributes

With everything discussed above, it may result in some changes for the final implementation; Jobinator may have other additions such as:

Reserved

- PDF or other popular file types with export utilities
- Logging tools or analytics for monitoring performance and usage
- Authentication services
- Database managing systems for efficiency
- For refined answers and reasoning, AI or natural language processing services
- URL handling and other tools that would handle analyzing user-provided references or uploaded documents

These parts/components help the system provide efficiency to the customer/user. Providing reusability and ease of access when it comes to using it as a proper tool.

2.3 Design and Implementation Constraints

1. Platform and Tech Constraints

The project should remain compatible with the technologies selected for implementation. The system is built as a browser-based application; design choices must support responsive front-end behavior, backend communication, and manageable integration with AI services.

2. Performance Constraints

Users should be able to interact with the system without excessive delay. Normal actions such as submitting information, receiving AI-generated suggestions, loading saved drafts, or updating a profile should be complete in a reasonable time.

Performance expectations should focus on usability rather than unrealistic speed goals. For example:

- Standard page interactions should respond quickly
- AI-generated revisions should be complete within an acceptable waiting period
- Saved user data should load reliably without repeated refresh attempts

The system should also be designed so that regular user growth does not immediately break main core functions.

Reserved

3. AI Output Limitations

The system depends on the whole “AI-generated” suggestion to improve resumes, CVs, and LinkedIn sections. Because AI-generated text is not always perfectly accurate or context-aware, the platform cannot assume that every result will be factually correct, role-appropriate, or professionally ideal without user review. We want to completely avoid any hallucinations, mistakes, or cyber-psychosis.
So it is best that Jobinator should be designed as a recommendation and drafting assistant rather than a fully autonomous decision-maker.

4. Security and Privacy Constraints

Jobinator handles career-related user data, which may include personally identifiable information such as names, contact information, education history, and employment history. For that reason, the system should protect stored and transmitted data/information through basic security practices such as:

- Protected user sessions
- Secure authentication
- Encrypted communication over HTTPS servers on websites
- Restricted access to stored user documents

With even all this, the system should also avoid collecting unnecessary personal data. Any plans for **ANY** kind of AI-processing workflow should be designed with privacy awareness first, especially if third-party services are used for text generation or document analysis.

5. Legal and Ethical Constraints

The system needs to avoid presenting false credentials, fabricated experiences, or misleading claims as acceptable outputs. Jobinator may improve wording and presentation; it shouldn’t be used as a tool for creating deceptive job applications.
At the moment the platform supports imported documents or profile data/content; it must also respect ownership and privacy boundaries related to users who submit files and use third-party platform content.

Reserved

6. UI/UX Constraints

The interface should remain simple enough for first-time job seekers, students, and general users who may not be experienced with professional document writing. This will mean that the design should have:

- Clean navigation
- Simple and straightforward editing
- Structure that is easy to read
- AI suggestions that are easy to read and understand
- Visible sections when labeled

The whole platform will not bury important actions under too many layers of menus.

7. Development Process Constraints

Development is constrained by course deadlines, the team skill level, available installment times, and organizing and formulating documentation alignment with the physical project itself, diagrams, and prototypes for testing beta builds.

2.4 Assumptions and Dependancies

The following assumptions and dependencies describe conditions that Jobinator depends on for normal functional operation, along with risks that could affect system behavior if those conditions are not met.

1. AI Output Accuracy and Relevance

Assumption: The AI service used by Jobinator will generally return relevant, structured, and professionally useful content based on the user's input.

Risk: AI generated text may be inaccurate, too generic, poorly phrased, or not fully appropriate for a specific job role or document section. This could reduce the quality of generated resumes, CVs, LinkedIn content, or job tailored revisions.

Affected Requirements:

- Functional Requirement 3.1 (AI content generation)
- Functional Requirement 3.2 (AI assisted rewriting)
- Functional Requirement 3.3 (Stronger wording suggestion)

Reserved

- ❖ Functional Requirement 4.3 (Job specific keyword and wording refinement)
- ❖ Functional Requirement 6.1 – 6.3 (document analysis and suggestions)
- ❖ Quality Attribute 6.2.3 (AI suggestions must be understandable and reviewable)

2. External AI Service Availability

Dependency: Jobinator depends on an external AI or large language model service for generation, rewriting, refinement, and suggestion based output.

Risk: If the service is unavailable, rate-limited, delayed, or returns unstable responses, core features such as content generation, rewriting, and job tailoring may become slower, limited, or temporarily unavailable.

Affected Requirements:

- ❖ Functional Requirement 3.1 – 3.3 (AI generation and improvement)
- ❖ Functional Requirement 4.3 (AI assisted job tailoring)
- ❖ Functional Requirement 7.5 (Admin AI configuration)
- ❖ Quality Attribute 6.1.2 (Acceptable AI response time)
- ❖ Quality Attribute 6.3.1 (reliable availability)
- ❖ Quality Attribute 6.5.1 – 6.5.3 (Scalability and future AI expansion)

3. User Input Completeness and Accuracy

Assumption: Users will provide truthful, relevant, and reasonably complete personal, academic, and professional information when creating or improving a document.

Risk: If the user provides incomplete, vague, outdated, or inaccurate information, the system may generate weak, misleading, or incomplete output. The quality of AI suggestions depends heavily on the quality of the input provided.

Affected Requirements:

- ❖ Functional Requirement 2.3 (Manual information input)
- ❖ Functional Requirement 2.6 (guided prompts for missing information)
- ❖ Functional Requirements 4.1 – 4.4 (Job specific tailoring)
- ❖ Functional Requirements 6.1 – 6.3 (Missing/weak section detection and improvement prompts)
- ❖ Quality Attribute 6.2.4 (guided structure for first time users)

Reserved

4. File Upload and Reference Processing

Dependency: Jobinator depends on document processing tools, file handling utilities, and reference/URL processing support for uploaded resumes, CVs, pasted content, and LinkedIn profile references.

Risk: Uploaded files may fail to parse correctly, formatting may be lost, pasted content may be inconsistent, or LinkedIn profile references may not provide usable data for refinement. This can reduce the quality of document analysis or interrupt the input workflow.

Affected Requirements:

- Functional Requirement 2.4 (Upload or paste existing resume/CV)
- Functional Requirement 2.5 (LinkedIn content or URL input)
- Functional Requirement 2.6 (Guided prompts after input)
- Functional Requirement 5.4 (Document Export)
- Software Interface 5.2.4 (Document processing tools)
- Software Interface 5.2.6 (reference and URL handling)
- Quality Attribute 6.3.3 (Retry after failed upload or AI processing)

5. Privacy, Security, and Permission Handling

Assumption: Users expect their document data, account data, and upload materials to be stored and processed securely, with clear permission controls in place.

Risk: Unauthorized access, weak session handling, unclear permission requests, or improper handling of uploaded content could expose sensitive information and reduce trust in the platform.

Affected Requirements:

- Functional Requirement 7.4 (Admin access restriction)
- Functional Requirements 8.1 – 8.5 (Privacy and Permissions)
- Communications Interface 5.3.1 (HTTPS Communication)
- Communications Interface 5.3.6 (Authentication and session communication)
- Quality Attributes 6.4.1 – 6.4.5 (Security protections)

6. System Performance Under Load

Dependency: Jobinator depends on adequate backend resources, storage access, and network stability to support authentication, uploads, AI requests, draft saving, and export actions.

Reserved

Risk: If too many users interact with the system at the same time, or if the backend is under provisioned, users may experience slower responses, failed saves, delayed AI output, or interrupted export workflows.

Affected Requirements:

- Quality Attribute 6.1.1 – 6.1.3 (Performance)
- Quality Attribute 6.3.1 – 6.3.4 (Reliability)
- Quality Attribute 6.5.1 – 6.5.2 (Scalability)
- Functional Requirements 5.1 – 5.4 (Saving, editing, and exporting)

7. Browser, Device, and Interface Compatibilty

Assumption: Users will access Jobinator through supported browsers, modern devices, and a stable browser based environment.

Risk: Differences between browsers, screen sizes, device limitations, or unsupported environments may cause layout issues, inconsistent interface behavior, broken uploads, or weaker accessibility support.

Affected Requirements:

- User Interface 5.1.1 – 5.1.8
- Quality Attribute 6.2.1 – 6.2.4 (Usability)
- Quality Attribute 6.7.1 – 6.7.4 (Accessibility)
- Quality Attribute 6.8.1 – 6.8.4 (Compatibility)

8. Data Persistence and Export Reliability

Dependency: Jobinator depends on stable storage systems and export utilities so that drafts, edits, and finished documents are preserved correctly.

Risk: If data is not saved properly, if export tools fail, or if document formatting breaks during output, users may lose progress or receive incomplete final files.

Affected Requirements:

- Functional Requirement 5.1 (Saved drafts)
- Functional Requirement 5.2 (Reopen drafts)
- Functional Requirement 5.3 (Manual editing before finalization)
- Functional Requirement 5.4 (Export to PDF or DOCX)
- Quality Attributes 6.3.2 – 6.3.3 (Preserve data and recover from failure)
- Quality Attribute 6.8.4 (Usable exported file formats)

Reserved

3. System Features

3.1 Functional Requirements

1. User Account Management

(FR = Functional Requirements)

FR1.1: The system shall/will allow users to create a personal account using required registration information.

FR1.2: The system shall/will allow registered users to log in and log out securely.

FR1.3: The system shall/will allow users to update account information, profile settings, and basic preferences.

FR1.4: The system shall/will allow users to recover or reset account access through an authentication-supported method.

2. Document Creation and Input Management

(FR = Functional Requirements)

FR2.1: The system shall/will allow users to choose whether they want to work on a resume, CV, or LinkedIn profile.

FR2.2: The system shall/will allow users to choose whether they want to create a new document from scratch or improve an existing one.

FR2.3: The system shall/will allow users to manually enter personal, academic, and professional information into the platform.

FR2.4: The system shall/will allow users to upload or paste an existing resume or CV for review and any improvements.

FR2.5: The system shall/will allow users to provide existing LinkedIn profile content through manual input or a LinkedIn profile URL for analyzing to help the AI construct or improve parts of it.

FR2.6: The system shall/will provide guided/directed prompts or question-based input fields to help the user supply missing information needed to build or improve a document.

Reserved

3. AI Content Generation and Improvements

(FR = Functional Requirements)

FR3.1: The system shall/will use an AI service to generate resume, CV, or LinkedIn profile content based on user provided information.

FR3.2: The system shall/will allow users to request AI assisted rewriting of existing sections to improve the clarity, the structure, and the professionalism that is needed for such documents.

FR3.3: The system shall/will allow the AI service to suggest stronger wording for the summaries, the job descriptions, the skills, and achievement statements.

FR3.4: The system shall/will give/present AI generated content as suggestions or something that can be editable output rather than final locked content

4. Job-Specific Tailoring

(FR = Functional Requirements)

FR4.1: The system shall/will allow users to provide a target job title, position, or industry for document refinement.

FR4.2: The system shall/will allow users to provide a job description so the document can be tailored more closely to a designated/desired role.

FR4.3: The system shall/will use AI assistance to adjust the wording, the emphasis, and relevant proper keywords based on the targeted role that was assigned or the uploaded job descriptions.

FR4.4: The system shall/will contain and preserve the user's original qualifications while tailoring content to remain truthful and relevant to the role.

5. Document Saving, Editing, and Exporting

(FR = Functional Requirements)

FR5.1: The system shall allow users to save incomplete or completed drafts of resumes, CVs, and LinkedIn profile content.

FR5.2: The system shall/will allow users to reopen and continue editing previously saved drafts.

FR5.3: The system shall/will allow users to manually edit AI generated or AI improved content before finalizing it.

FR5.4: The system shall/will allow users to export completed documents in a commonly used file format such as PDF or DOCX.

Reserved

6. Feedback and Suggestions

(FR = Functional Requirements)

FR6.1: The system shall/will analyze user documents and identify missing or weak sections such as summaries, skills, experience descriptions, or education details.

FR6.2: The system shall/will provide suggestions for improving the wording, formatting, or completeness of the document.

FR6.3: The system shall/will notify users when more information may be needed to generate a stronger or more accurate document.

FR6.4: The system shall/will allow users to provide feedback on AI generated suggestions so later recommendations can be improved.

7. Admin Monitoring and Management

(FR = Functional Requirements)

FR7.1: The system shall/will allow authorized administrators to view and manage user accounts when necessary.

FR7.2: The system shall/will allow administrators to monitor general system activity, usage amount and status, and document process/processing behaviors.

FR7.3: The system shall/will allow administrators to review reported issues, system errors, or failed requests involving uploads, AI suggestions, or saved drafts. Any errors that require an administrator's fix

FR7.4: This system shall/should restrict administrative functions to authorized admin access only.

FR7.5: The system shall/will allow authorized administrators to configure or manage AI service settings, model connections, or related integration behavior when necessary.

8. Privacy and Permissions

(FR = Functional Requirements)

FR8.1: The system shall/will request user permission before storing/analyzing/revising uploaded documents or profile content.

FR8.2: The system shall/will allow the user to control whether their information is used for AI assisted analysis and document improvements. (On by default).

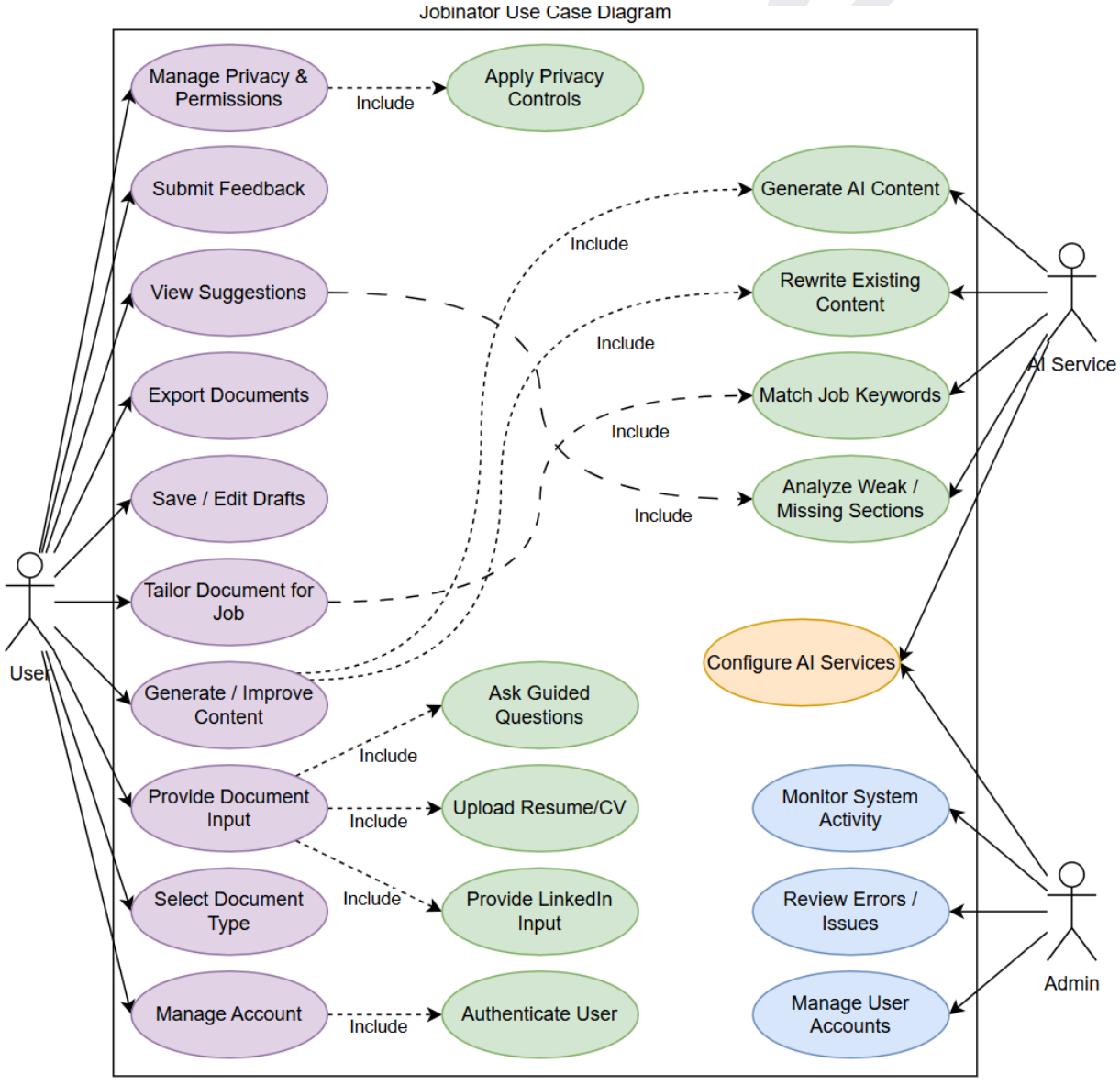
FR8.3: The system shall/will protect user data through authenticated access and secure session handling.

Reserved

FR8.4: The system shall/will allow the user to delete saved drafts, uploaded files, or account related document data from the platform.

FR8.5: The system shall/will avoid collecting unnecessary personal data beyond what is needed for document creation, improvement, and account support.

3.2 Use Case Diagram



3.3 Use Case Description

Actors:

- **User**
- **Admin**
- **AI Service (external actor)**

User:

1. **Manage Account**
   A. **Main Actor:** User
   B. **Description:** Allows the users to create an account, log in, log out, update their profile information, and recover account access when needed.
   C. **Precondition:** User has access to the Jobinator platform.

2. **Select Document Type**
   A. **Main Actor:** User
   B. **Description:** Allows the user to choose whether they want to work on a resume, CV, or LinkedIn profile.
   C. **Precondition:** User is logged into the system or has access to the document setup interface.

3. **Provide Document Information**
   A. **Main Actor:** User
   B. **Description:** Allows the user to enter information manually, upload an existing resume/CV, or provide LinkedIn content or a LinkedIn profile URL.
   C. **Precondition:** User has selected a document type.

4. **Generate / Improve Content**
   A. **Main Actor:** User
   B. **Description:** Allows the user to generate new content from scratch or improve existing content using AI assisted suggestions or recommendations and rewriting tools.
   C. **Precondition:** User has provided enough document information for processing.

Reserved

5. **Tailor Document for Job**
    A. **Main Actor:** User
    B. **Description:** Allows the user to provide a job title, position, industry, or job description so the system can refine the document towards the targeted role.
    C. **Precondition:** User has an existing or generated document and is available for refinement.

6. **Save / Edit Drafts**
    A. **Main Actor:** User
    B. **Description:** Allows the user to save document drafts, reopen them later, and continue editing or revising what it was given.
    C. **Precondition:** User is authenticated and has active documents that are in progress.
7. **Export Documents**
    A. **Main Actor:** User
    B. **Description:** Allows the user to export completed documents in commonly used formats such as PDFs.
    C. **Precondition:** User has a finished or almost finished document ready for export.

8. **View the Suggestions**
    A. **Main Actor:** User
    B. **Description:** Allows the user to review AI generated suggestions related to weak wording, any missing sections, the format, or completed documents.
    C. **Precondition:** The system has analyzed the user's document or content input.

9. **Submit Feedback or Comments**
    A. **Main Actor:** User
    B. **Description:** Allows the user to provide feedback on AI generated suggestions so future recommendations can be improved.
    C. **Precondition:** User has received and reviewed AI suggestions or rewritten content.

10. **Manage Privacy & Permissions**
    A. **Main Actor:** User
    B. **Description:** Allows the user to control privacy settings, approve or deny AI assisted analysis, and manage stored files or document related data/information that is tied to the account when created.
    C. **Precondition:** User is authenticated and has access to account or document settings.

Reserved

Admin:

11. Monitor System Activity

A. **Main Actor:** Admin
B. **Description:** Allows the administrator to monitor the general system activity, usage status, and document processing behavior across the platform.
C. **Precondition:** Admin is authenticated and has proper authorization.

12. Review Errors / Issues

A. **Main Actor:** Admin
B. **Description:** Allows the administrator to review reported issues, failed uploads, AI processing problems, or saved draft errors that may require the system's attention.
C. **Precondition:** Admin is authenticated and system logs or error reports are available.

13. Manage User Accounts

A. **Main Actor:** Admin
B. **Description:** Allows the administrator to view, manage, or restrict user accounts when necessary for platform maintenance or support.
C. **Precondition:** Admin authorization is active.

14. Configure AI Services

A. **Main Actor:** Admin
B. **Description:** Allows the administrator to configure or manage AI service settings, shape or form connections while they are managed, or related usage behavior used by Jobinator.
C. **Precondition:** Admin is authenticated and AI integration is enabled in the system.

AI Service (External System)

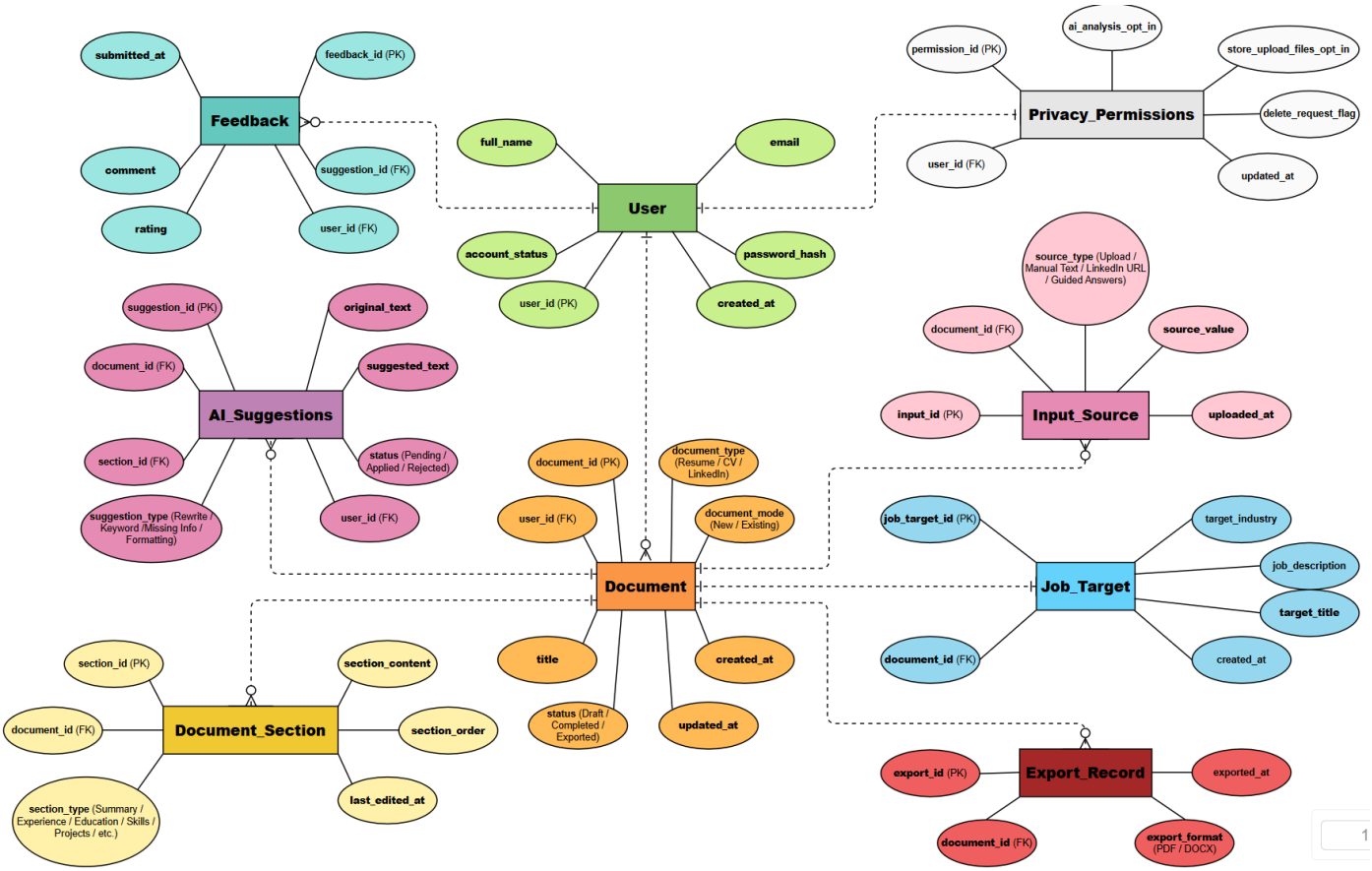
Description:

Serves as the external AI processing component used by Jobinator to generate new content, rewrite existing sections, match any job related keywords, and produce any suggestions for improvement.

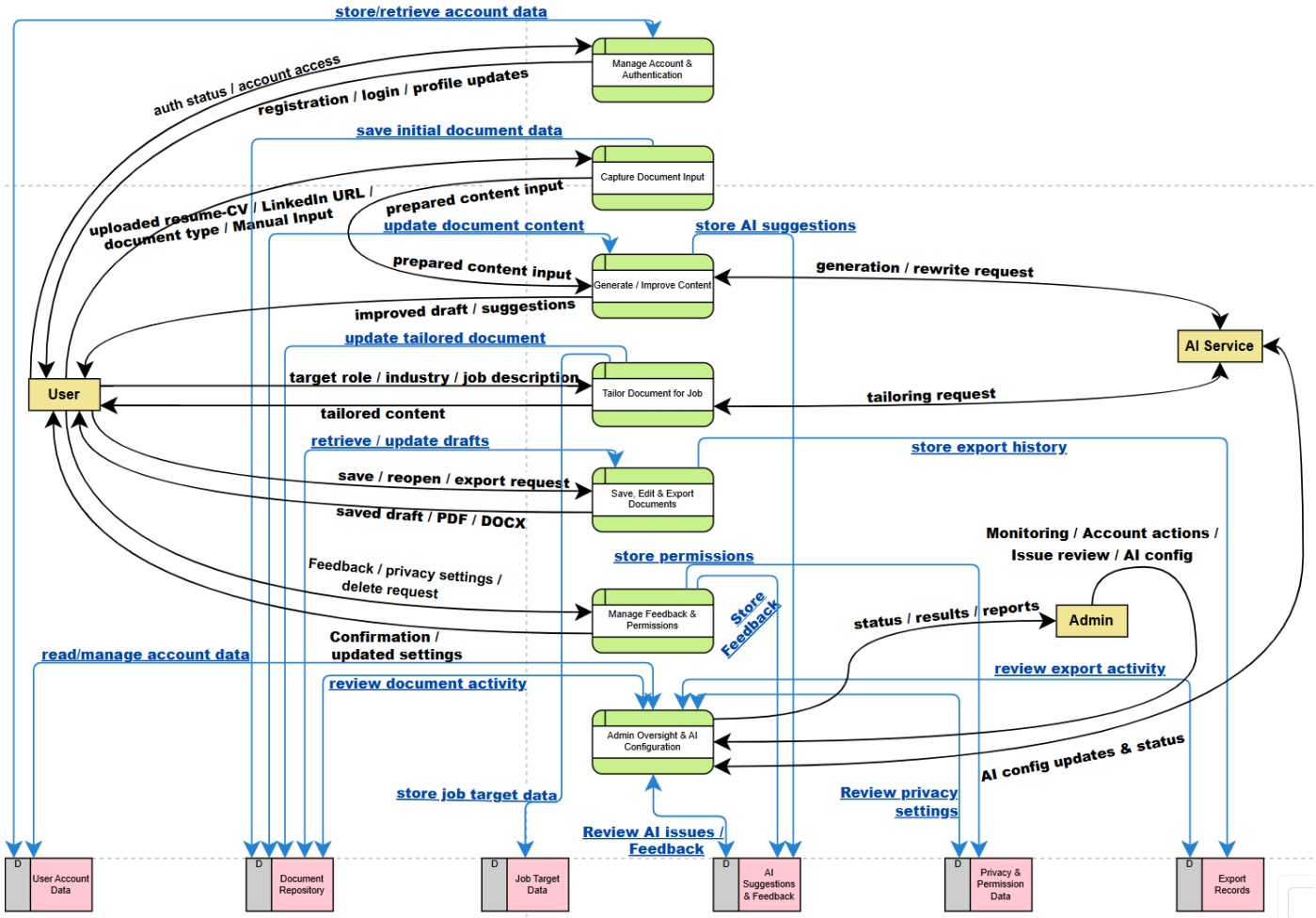
Reserved

4. Data Requirements

4.1 Entity Relationship Diagram



4.2 Data Flow Diagram



4.3 Mockup Design

The following mockups represent the user interface design of Jobinator. These screens are aligned with the system's functional requirements, including user authentication, document creation, document input, AI assisted editing, job specific tailoring, and document export. The design is intended to remain simple, guided, and usable across both desktop and mobile platforms.

1. Login

The login screen provides a secure and simple entry point for users accessing the platform. The interface is designed with a clean and minimal layout to reduce confusion for first time users.

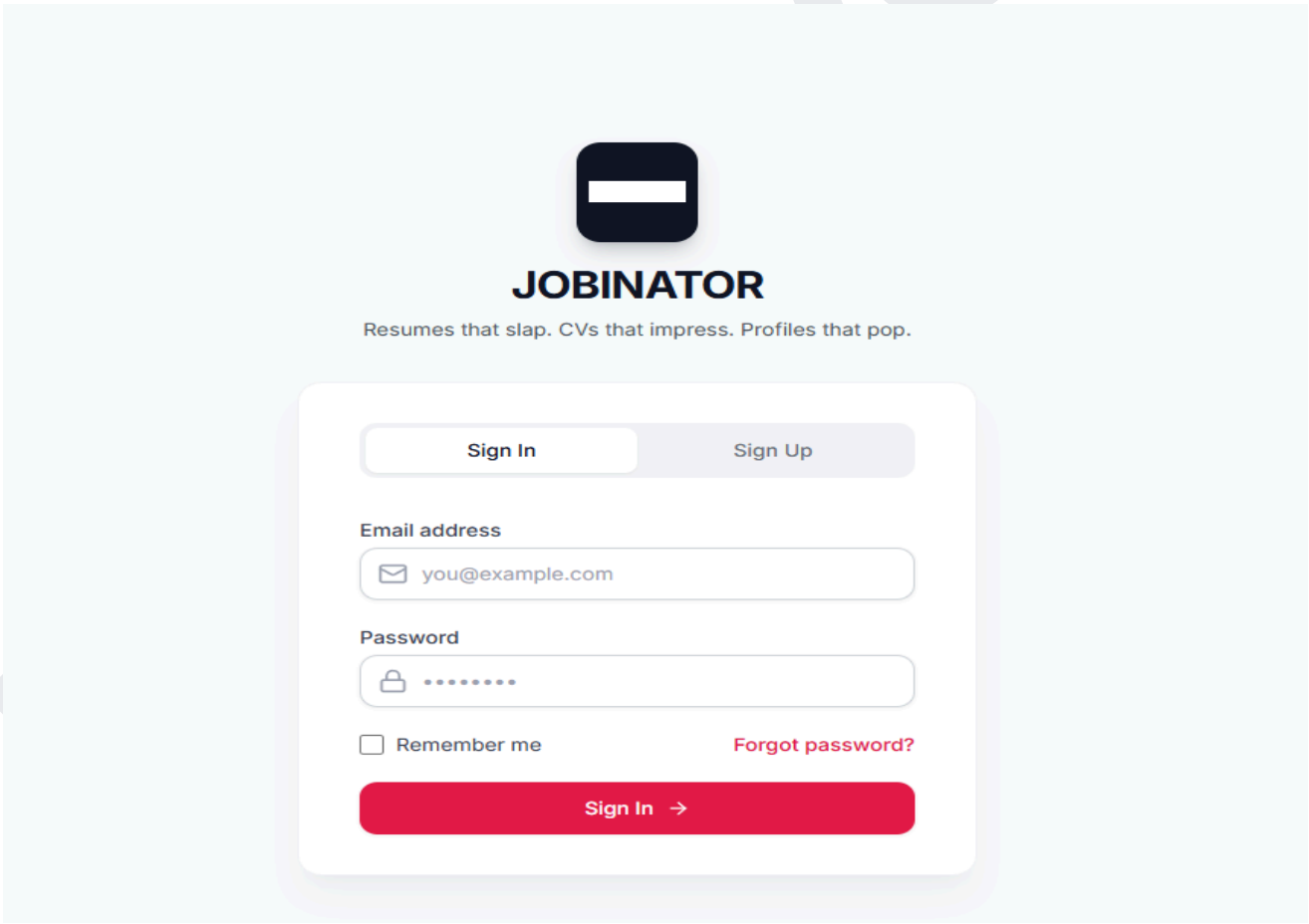
Key Interface Elements:

- Sign in and account access
- “Remember me” option for easier repeated access
- “Forgot password” option for account recovery
- Clear path to account creation for new users

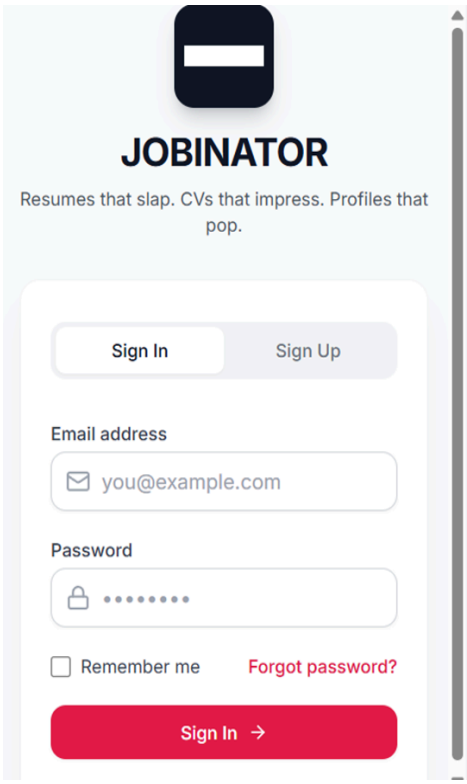
Supported Requirements:

- FR1.1–FR1.4

Desktop view:



Mobile View:



2. Dashboard View:

The dashboard serves as the user's main workspace after logging in. It allows users to manage ongoing work, reopen saved drafts, and quickly start a new document.

Key Interface Elements:

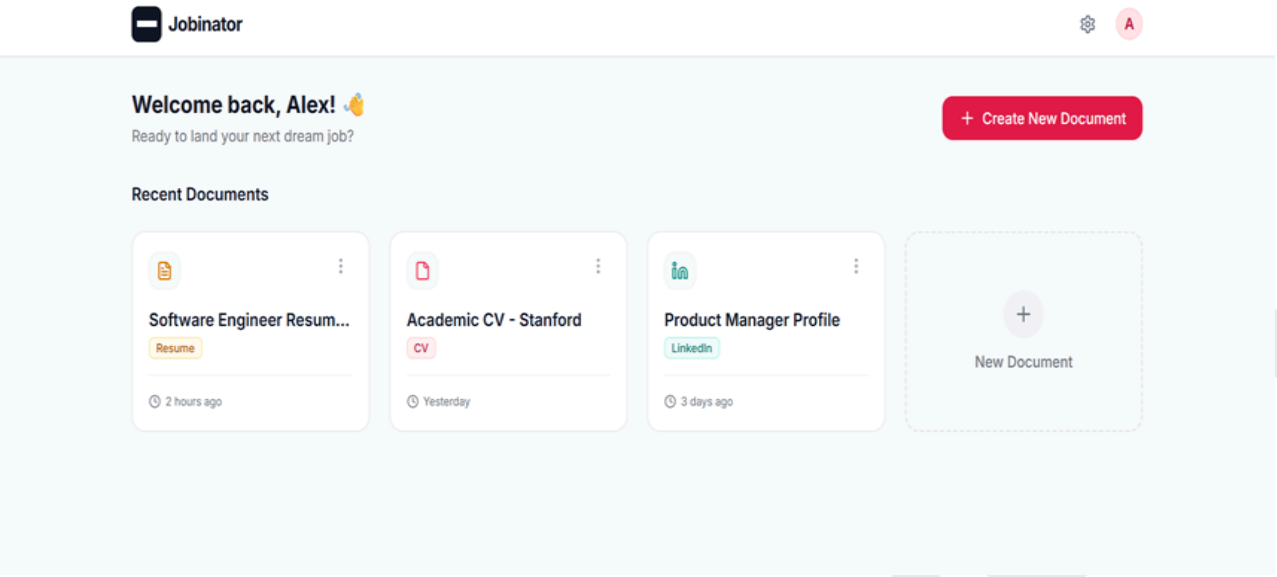
- Saved drafts and recently edited documents
- Clear "Create New Document" action
- Simple navigation for continuing existing work
- Organized layout for efficient workflow continuation

Supported Requirements:

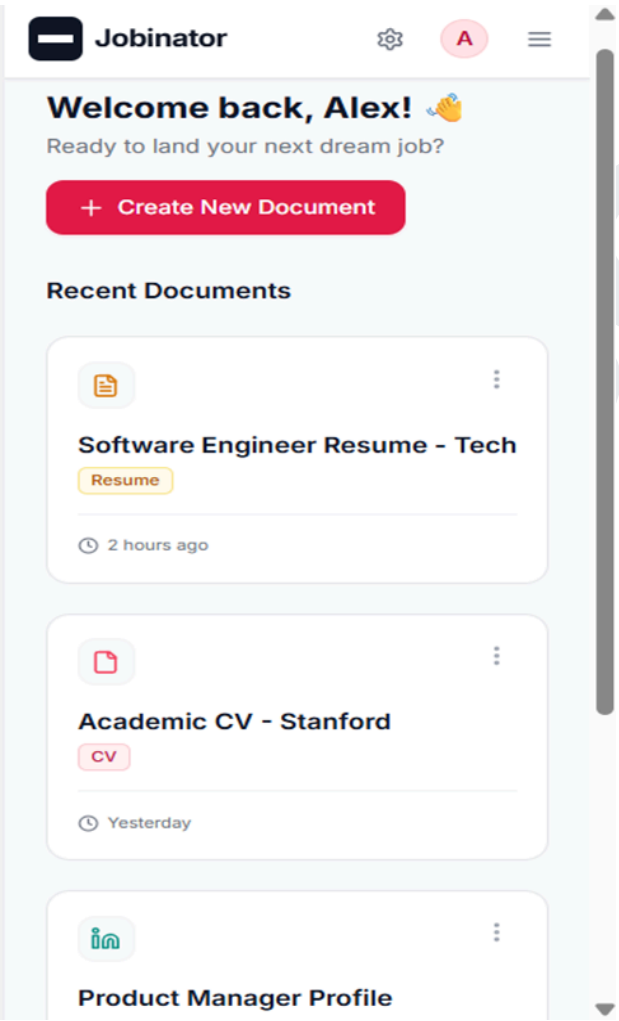
- FR5.1–FR5.2

Reserved

Desktop view:



Mobile View:



3. Create View

This screen allows users to begin a new workflow by choosing the type of career document they want to work on and whether they want to create it from scratch or improve an existing version.

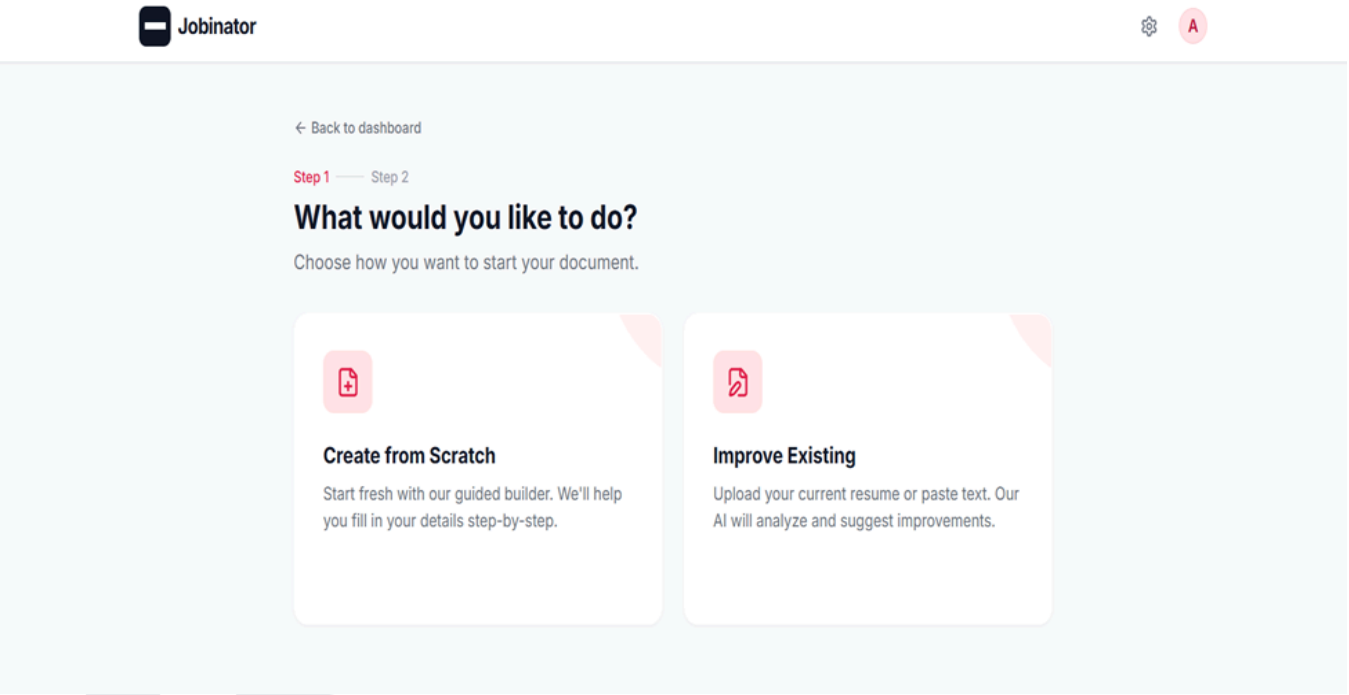
Key Interface Elements:

- Select **Resume**, **CV**, or LinkedIn Profile
- Choose **Create New** or **Improve Existing**
- Guided progression into the next input step
- Beginner-friendly structure for first time users

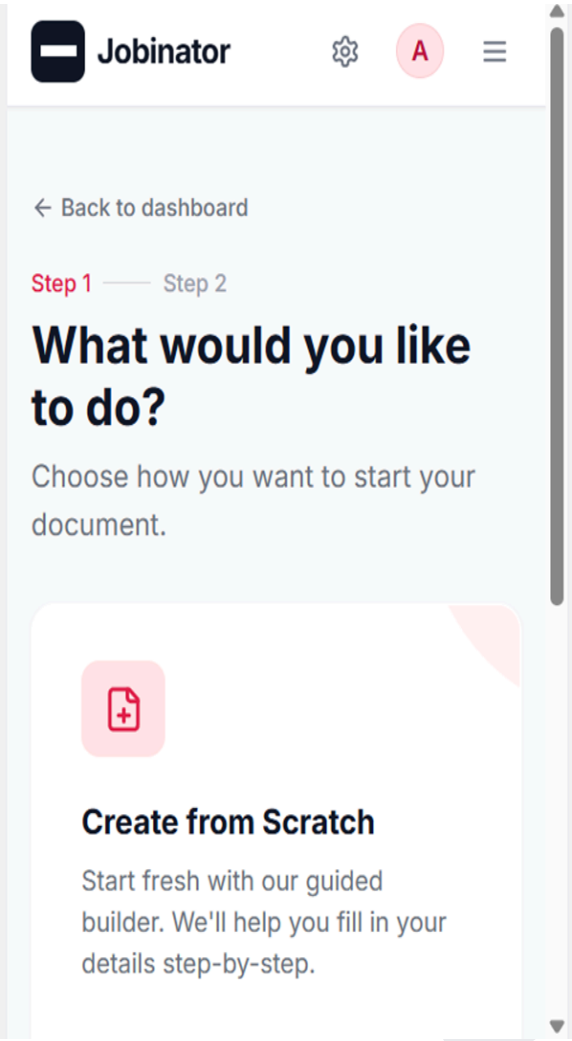
Supported Requirements:

- FR2.1–FR2.2

Desktop View:



Mobile View:



4. Input View

The input screen allows users to submit the information that Jobinator will use for analysis and improvement. This supports both manual entry and existing-document workflows.

Key Interface Elements:

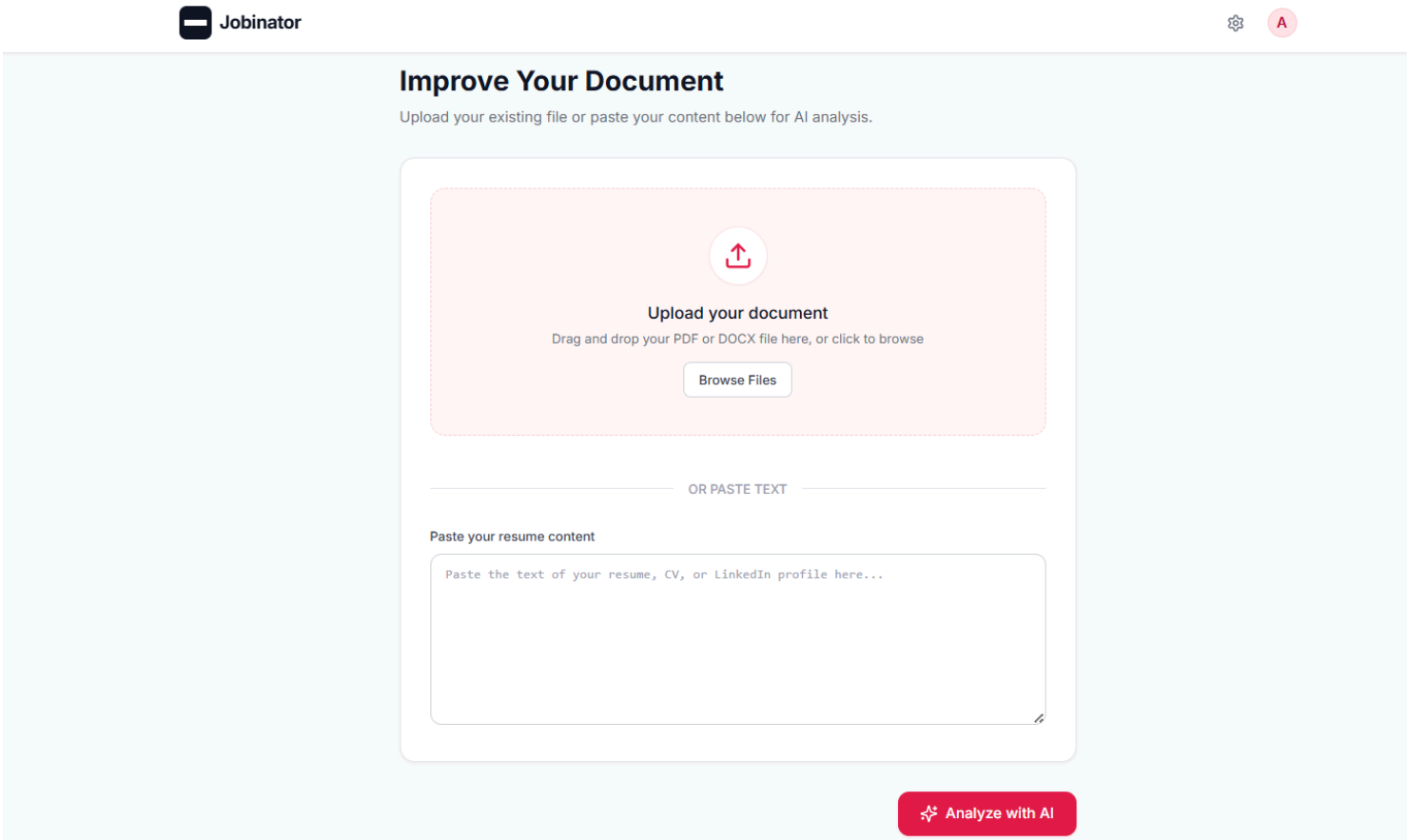
- Drag and drop upload area for resume or CV files
- Text box for pasting existing content manually
- LinkedIn profile input through pasted text or profile URL
- Guided prompts or question based fields for missing information
- AI analysis trigger for content review and preparation

Supported Requirements:

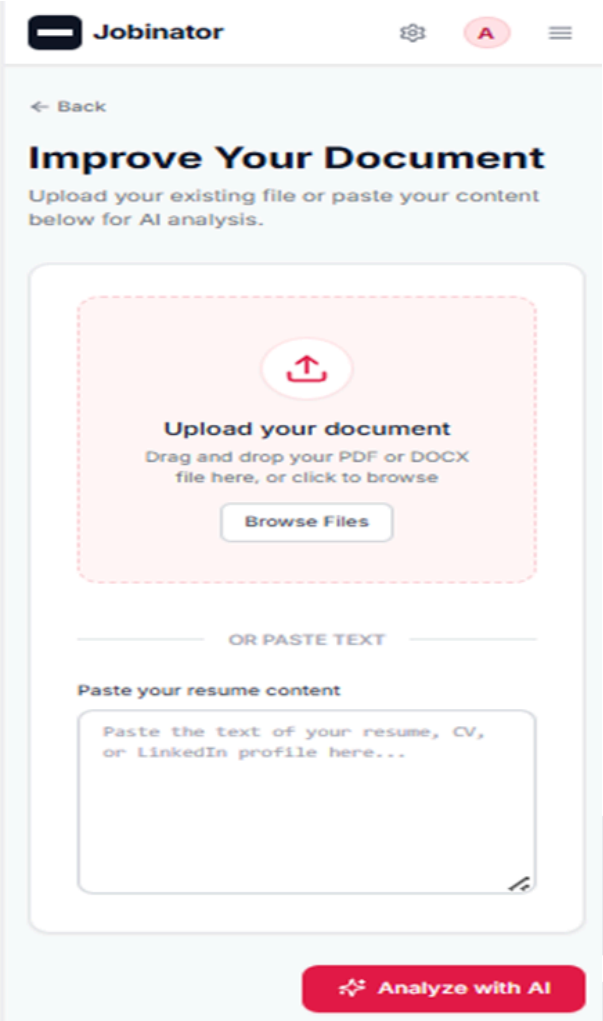
- FR2.3–FR2.6

Reserved

Desktop View:



Mobile View:



5. Editor with AI Suggestions View

This is the main editing workspace where users review, edit, and improve their content with AI assisted help. The interface should make AI suggestions visible without taking control away from the user.

Key Interface Elements:

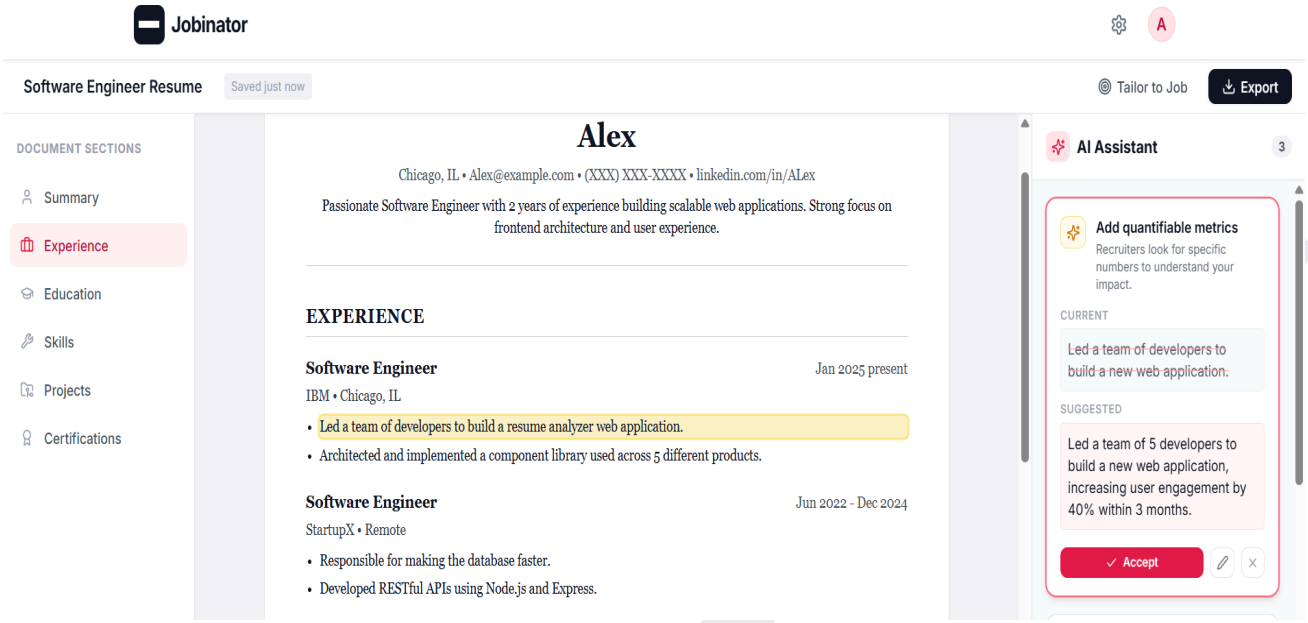
- Navigation between document sections such as Summary, Experience, Education, Skills, and Projects
- Main editable document area
- AI suggestions panel showing improved wording, missing skills, rewritten bullet points, and clarity improvements
- Editable AI generated output rather than locked final text
- Visual prompts for weak or incomplete sections

Reserved

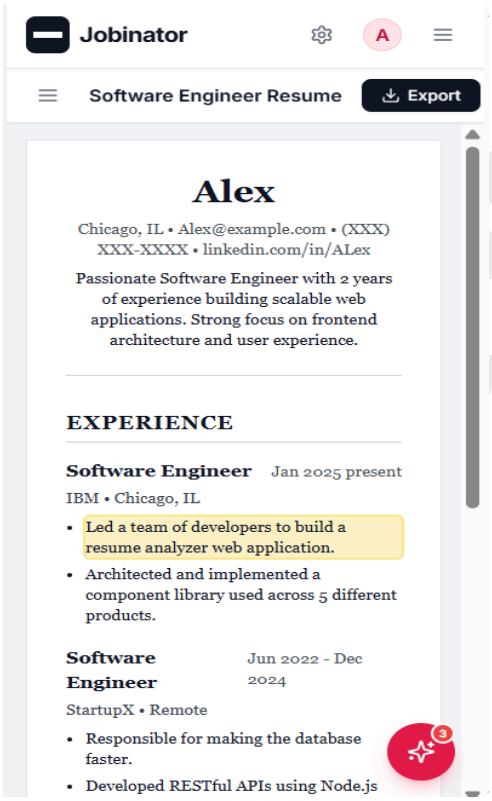
Supported Requirements:

- FR3.1–FR3.4 and FR6.1–FR6.4

Desktop View:



Mobile View:



6. Tailor Document View

This screen allows users to tailor their document toward a specific job role. It supports targeted refinement based on a job title, industry, or full job description.

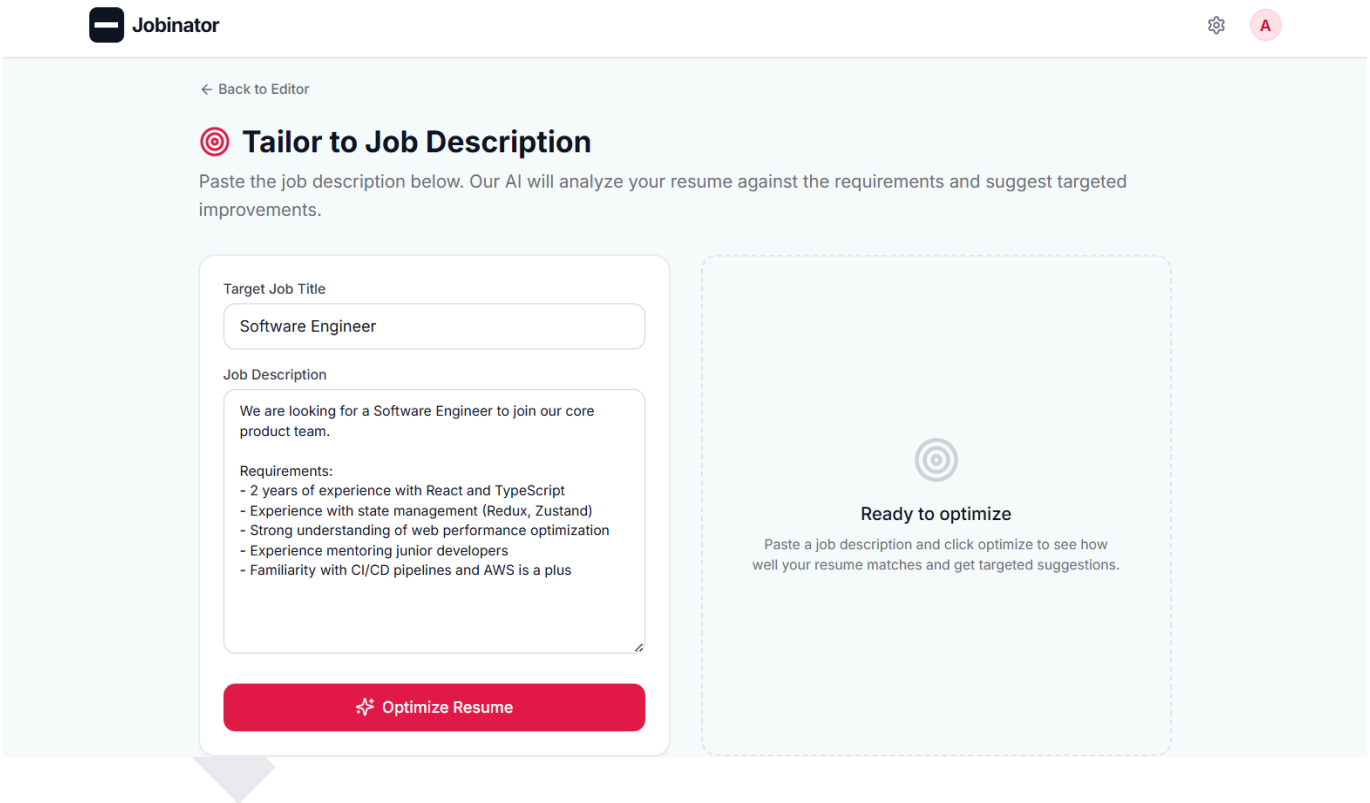
Key Interface Elements:

- Input field for target job title, role, or industry
- Text area for pasted job description
- AI optimization button for targetted refinement
- Suggestions that adjust wording, emphasis on parts, and keywords to better match the selected role
- Preservation of the user's original qualifications while improving relevance

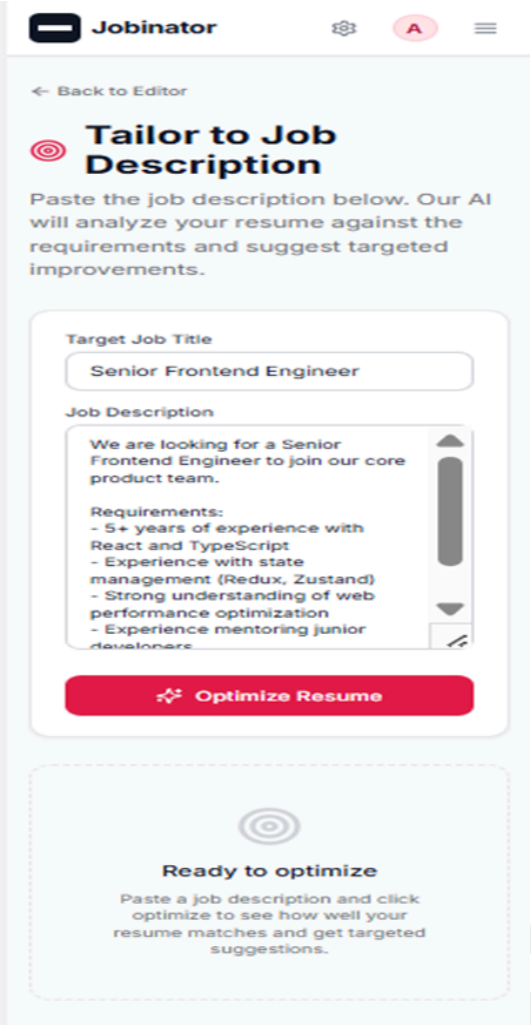
Supported Requirements:

- FR4.1–FR4.4

Desktop View:



Mobile View:



7. Export View

The export screen allows the user to review the completed document and download it in a commonly used format for outside use.

Key Interface Elements:

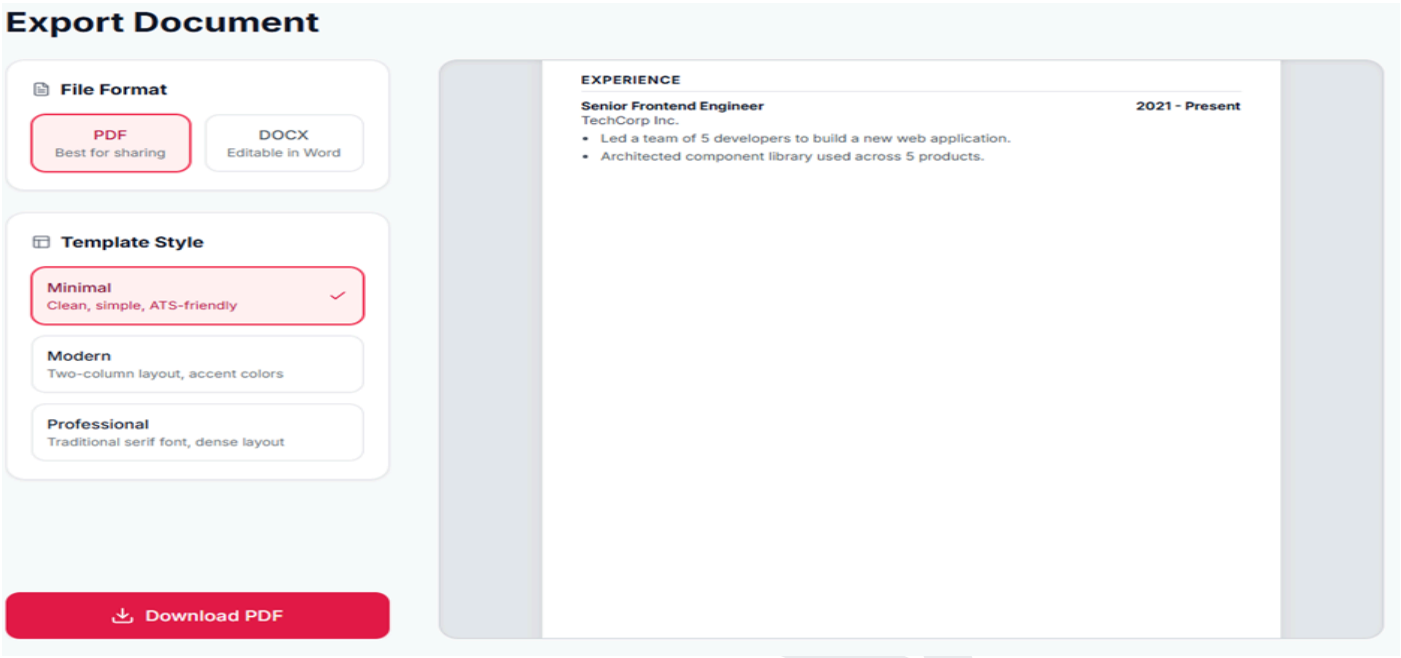
- Final document preview
- Export/download button
- Format selection such as **PDF** or **DOCX**
- Clean layout for final review before export

Supported Requirements:

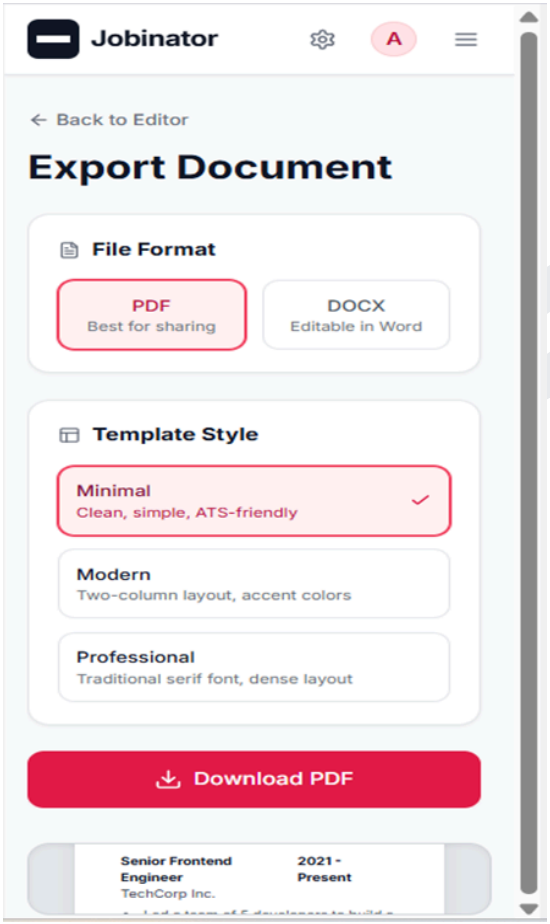
- FR5.4

Reserved

Desktop View:



Mobile View:



5. External Interface Requirements

5.1 User Interface

1. Web-Based User Interface:

Jobinator will provide a browser based user interface that allows users to access the platform without requiring special installed software. The interface will support the system's main view, including login, dashboard, document creation, document input, AI assisted editing, job specific tailoring, and export functions.

2. Responsive Design:

The interface will adapt to desktop, laptop, tablet, and mobile screen sizes. Layouts should remain readable and usable across different display sizes so users can interact with Jobinator without major visual or navigation issues.

3. Document Input Interface:

The system will provide input forms and upload areas that allow users to manually enter personal, academic, and professional information; upload an existing resume or CV; or provide LinkedIn profile content or a LinkedIn profile URL for AI analysis.

4. Guided Question Interface:

Jobinator will provide guided prompts and questions to help users fill in missing information when building or improving a document. This interface is intended to reduce confusion for first time users and improve the completeness of submitted content.

5. Document Editor Interface:

The platform will provide an editable workspace where users can view and revise document sections such as summary, experience, education, skills, and projects. AI generated suggestions should appear in a visible, organized manner, allowing the user to maintain control over the final part of their edits/content.

Reserved

6. Tailoring Interface:

The system will provide a dedicated interface for job specific tailoring, where users can enter a target job title, role, industry, or job description. This view will support AI assisted refinement of wording, emphasis, and keywords.

7. Export Interface:

Jobinator will provide a final review and export interface that allows the user to preview completed documents and export them in supported formats such as PDF and/or DOCX.

8. Administrative Interface:

The system will provide an administrative interface for authorized users to monitor system activity, review issues, manage user accounts, and configure AI service settings when necessary.

5.2 Software Interfaces

1. AI Service Interface:

Jobinator has an interface with an external AI or large language model service to generate new document content, rewrite existing sections, match job related keywords, and produce improvement suggestions.

2. Backend Application Server:

The system will interface with a backend application server responsible for processing user requests, handling session management, coordinating document workflows, and routing data between the frontend, storage systems, and AI services.

3. Database and Storage Interface:

Jobinator will interface with a structured database or managed cloud storage system to store user accounts, saved drafts, uploaded content, document sections, job target information, AI generated suggestions, privacy settings, and export records.

Reserved

4. Document Processing Interface:

The system will interface with document processing tools or libraries capable of handling uploaded resume/CVs and exporting completed documents in commonly used file formats such as PDF and DOCX.

5. Authentication Interface:

Jobinator will interface with authentication and session handling services to support account creation, secure login, password recovery, profile updates, and protected access to saved user data.

6. Reference and URL Handling Interface:

The system will interface with tools that can process user provided LinkedIn profile references, pasted text, profile URLs, and job description input for AI assisted analysis and refinement.

7. Administrative AI Configuration Interface:

The system will interface with configurable AI service settings so authorized administrators can manage model connections or related integration behavior when it is required.

5.3 Communications Interface

1. HTTPS Communication:

All communication between the client interface, backend server, and connected services should occur over secure HTTPS connections to protect sensitive user information.

2. RESTful API Communication:

Jobinator will use API based communication between the frontend and backend systems, likely using standard HTTPS methods such as GET, POST, PUT, and DELETE to support account actions and document processing, saving, and export requests.

Reserved

3. JSON Data Exchange:

Structured data exchanged between the frontend, backend, and AI service should use a lightweight format such as JSON to support efficient communication and system integration.

4. File Upload Communication:

The system will support secure file upload communication for resumes, CVs, and related document input. This allows uploaded files to be transferred from the user interface to the backend for storage and analysis.

5. AI Service Communication:

The backend will communicate with the external AI service through secured API requests in order to submit document content, job target data, rewriting requests, and refinement prompts and to receive generated suggestions in return.

6. Authentication and Session Communication:

The system will use secure communication methods for login credentials, session validation, password recovery, and protected account actions to reduce unauthorized access risks.

7. Admin Communication Pathways:

Administrative actions such as monitoring activity, reviewing issues, and configuring AI related behavior will communicate through protected internal system channels restricted to authorized administrators only.

Reserved

6. Quality Attributes

6.1 Performance

1.1: The system will respond to standard user actions such as login, page navigation, opening saved drafts, and saving document edits within a reasonable time under normal operating conditions.

1.2: AI assisted generation, rewriting, and job tailoring requests should return results within an acceptable waiting period during normal system use.

1.3: The system will process common upload, save, and export actions without excessive delay, excluding delays caused by network conditions, unusually large files, or third party AI service response times.

6.2 Usability

2.1: The interface will allow users to begin a new resume, CV, or LinkedIn workflow in a clear and guided manner without requiring advanced technical knowledge.

2.2: Jobinator will maintain consistent navigation, labeling, and layout across its main screens, including login, dashboard, create, input, editor, tailor, and export views.

2.3: AI generated suggestions will be presented in a way that is easy to understand, review, and edit before the user finalizes any document content.

2.4: The platform will provide guided prompts, form inputs, and visual structure that reduces confusion for first time users and improves completion of mission document information.

6.3 Reliability

3.1: Jobinator should maintain reliable availability during normal use, excluding scheduled maintenance or outages caused by external service dependencies.

3.2: The system will preserve saved drafts and previously entered document data so users can return and continue their work without unnecessary loss of progression.

3.3: If an upload, save, or AI processing request fails, the system will provide a clear error response and allow the user to retry without deleting valid existing work.

Reserved

3.4: The system will handle ordinary document processing workflows in a stable manner without frequent crashes, broken sessions, or repeated refresh requirements.

6.4 Security

4.1: The system will use secure communication methods such as HTTPS and TLS to protect data transmitted between the client interface, backend services, and connected AI components.

4.2: Jobinator will enforce secure authentication and session handling practices to protect user accounts and stored document data.

4.3: Administrative functions will be restricted to authorized administrators only, and access to sensitive system controls will require proper authorization.

4.4: The system will request and respect user permission before storing, analyzing, or revising uploaded files or profile related content.

4.5: Jobinator will protect stored user information from unauthorized access and avoid collecting unnecessary personal data beyond what is needed for platform operation.

6.5 Scalability

5.1: The system architecture will be able to support growth in users, stored documents, and AI processing requests without requiring major redesign of core platform functions.

5.2: Jobinator should support increasing levels of concurrent activity while maintaining acceptable responsiveness for normal account, editing, and export workflows.

5.3: The system should be designed so future features, such as additional document types, stronger integrations, or expanded AI services, can be added without breaking the main platform structure.

Reserved

6.6 Maintainability

6.1: The codebase will be organized in a modular way so that the user interface, backend services, storage systems, and AI integrations can be updated more easily.

6.2: System components, workflows, and major integrations should be documented clearly enough to support debugging, testing, maintenance, and future development.

6.3: Updates, fixes, and refinements should be introduced with minimal disruption to active users whenever possible.

6.4: The platform should be designed so that interface changes, AI service adjustments, and document processing improvements can be made without rewriting the full system.

6.7 Accessibilty

7.1: Jobinator will support accessibility-friendly navigation features such as readable labeling, keyboard accessibility, and compatibility with common assistive technologies where possible.

7.2: The interface will use readable font sizes, organized layouts, and sufficient contrast to improve clarity across supported devices.

7.3: Important controls, status messages, and form inputs should be understandable to users with varying levels of experience and ability.

7.4: Visual elements, icons, and interface cues should be supported with text labels or accessible alternatives where they are appropriate.

6.8 Compatibility

8.1: The system will be compatible with major modern web browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

8.2: Jobinator will function across major operating systems and device types, including Windows, macOS, Linux, Android, and iOS, through responsive browser based access.

8.3: The platform will support common document workflows involving resumes, CVs, and LinkedIn related input across desktop and mobile environments.

8.4: Exported documents should remain usable in common file formats such as PDF and DOCX.

Reserved